MILITARY SPECIFICATION

BODY ASSEMBLY, TANK PERISCOPE: 10513391

1. SCOPE

1.1 This specification covers one type of periscope body assembly which is assembled to Head Assembly 10513392 to form the unity-power daylight, 9-power passive (low-light level) XM44 Periscope (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-O-52374	Oscillator; High-voltage

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F8565566	Pressure Tester-Two Station
F10513391	Body Assembly: 10513391

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DRAWINGS

U.S. Army Munitions Command

F10513500	Mount Assembly: 10513500
IEL 10513391	List of Inspection Documents for Body Assembly 10513391

PACKAGING DATA SHEET

MIL-P-14232/P10513391	Packaging of Body Assembly: 10513391

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The body assembly shall be manufactured in accordance with Drawing F10513391 and drawings pertaining thereto.

3.2 Image intensifier tube.- The image intensifier tube shall conform to the applicable purchase description (see 6.3).

3.3 High-voltage oscillator.- The high-voltage oscillator shall conform to MIL-O-52374.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the requirements of MIL-F-13926 specified below:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship
(f) Soldering electrical connections

3.5 Environmental.

3.5.1 Extreme storage temperatures.- The body assembly shall show no evidence of physical failure such as glass breakage and bonding or casting failure due to storage at temperatures ranging from -65° to +160° Fahrenheit (F). Subsequent to meeting this requirement, the body assembly shall be capable of meeting the requirements of 3.5.4.

3.5.2 Extreme operating temperatures.- The body assembly shall meet the applicable requirements of 3.19 while thermally stabilized at -40° and +150°F.

3.5.3 Standard temperature.- Unless otherwise specified herein, the body assembly shall meet the following requirements while exposed to a temperature between 60° and 90°F.

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3.5.4 Vibration.- The body assembly shall show no evidence of physical failure due to being vibrated at 30 cycles per second and an amplitude of 0.06 inch, 0.12 inch total excursion (see 3.12.1). Subsequent to meeting this requirement, the body assembly shall be capable of meeting the following requirements.

3.6 Operating voltages.- The operating voltage applied to the inlet connector of the reticle projector and image intensifier tube, shall be 5.5 ± 1.5 volts direct current (vdc).

3.7 Cleanliness.- The body assembly shall meet the cleanliness requirements of MIL-O-13830 and there shall be no foreign matter within the body assembly which interferes with observation or adversely affects image quality. The two viewing systems shall also meet the cleanliness requirements of 3.7.1 and 3.7.2.

3.7.1 Unity-power system.- There shall be no more than four particles of dirt visible on the optical surfaces of the unity-power system. The area of each particle shall not be in excess of an area equivalent to a size 10 dig.

3.7.2 Nine-power system.- There shall be no more than three particles of dirt visible when viewing through the eyepiece. The area of each particle shall not be in excess of an area equivalent to a size 10 dig.

3.8 Sealing.- The body assembly shall withstand an internal pressure of 5 ± 0.10 pounds per square inch gage (psig) with no evidence of leakage.

3.9 Dielectric strength.- Insulation resistance shall be not less than 10 megohms (mego) when a potential of 100 vdc is applied. This requirement shall be met with the image tube and power pack removed from the body assembly.

3.10 Image steadiness.

3.10.1 Flicker.- There shall be no visible flicker or movement of the image. This requirement shall be met subsequent to allowing a 15-second time lapse to permit stabilization of the tube.

3.10.2 Sparking.- There shall be no evidence of sparking. This requirement shall be met subsequent to allowing a 15 second time lapse to permit stabilization of the tube.

3.11 Projected reticle pattern.

3.11.1 Pattern definition.- The projected reticle pattern shall appear clearly defined and free of irregularities when observed through the eyepiece.

3.11.2 Reticle accuracy.- The angular value between the boresight cross and the maximum range graduation of the projected reticle pattern shall be within 2.0 percent of the value specified on the reticle drawing.

3.12 Collimation change (nine-power system).

3.12.1 Due to vibration.- An original line of sight shall not deviate more than 0.10 mil due to vibration (see 3.5.4).

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3.12.2 Collimation change (due to assembly to mount).- An original line of sight shall not deviate more than 0.10 mil due to repeated assembling of body assembly to the mount with which the body assembly is intended to be used.

3.13 Image tilt.

3.13.1 Unity-power system.- The image of a target line which is originally perpendicular to Datum Plane "B" shall emerge perpendicular to Datum Plane "A" within 1.0 degree of arc.

3.13.2 Nine-power system.- The image of a target line which is originally perpendicular to Datum Plane "B" shall emerge perpendicular to Datum Plane "A" within 1.0 degree of arc.

3.14 Parallelism of reticle and image.- The image of a target line which is originally parallel to Datum Plane "B" shall be parallel, within 30 minutes of arc, to a line defined by the image of the horizontal lines located nearest to the geometric axis of the reticle pattern (see applicable reticle drawing).

3.15 Boresight movements.- The requirements of 3.15.1, 3.15.2, and 3.15.3 shall be met by the azimuth and elevation boresight movements.

3.15.1 Excursion range.- This requirement shall be met with the boresight cross originally in coincidence with an incident line of sight which is:

(a) perpendicular to the intersection of Datum Planes "A" and "B".
(b) positioned to form a 91-degree 30 minute angle with respect to Datum Plane "A". This angle shall be as measured from Datum Plane "A" toward Datum Plane "B" in a clockwise direction as observed on Drawing F10513391, Sheet 2.

The boresight movements shall provide a minimum of 7.0 mils movement in each direction as read on the boresight scales.

3.15.2 Reticle shift.- Rotation of the boresight knobs in each direction, through the range specified in 3.15.1, shall not cause the reticle to deviate more than 0.20 mil in a direction perpendicular to the intended movement.

3.15.3 Backlash.- Backlash in each boresight movement shall not exceed 0.10 mil.

3.16 Resolution.

3.16.1 Unity-power system.- Resolution shall be 45 seconds of arc or less when scanning the exit aperture with a 10-power telescope having a 0.50 inch aperture.

3.16.2 Nine-power system.- Resolution for one best setting of tube focus shall not exceed "X" minutes. "X" shall be derived from the following:

$$X \text{ minutes} = \frac{23}{\ell \text{ -p/mm}}$$

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where ℓ-p/mm is the line pairs per millimeter at the photocathode surface of the image tube which can be resolved on the image screen. The ℓ-p/mm is specified in the applicable purchase description (see 6.3).

3.17 Dioptral adjustment.

3.17.1 Zero setting.- The diopter scale shall indicate zero within 0.25 diopter when the screen of the image tube is brought into focus using a diopto-meter set for zero diopters.

3.17.2 Excursion range.- The diopter movement shall be adjustable through a minimum range of -4.0 to + 4.0 diopters as read on the diopter scale.

3.17.3 Backlash.- Backlash in the diopter movement shall not exceed 1/8 diopter as read on the diopter scale.

3.18 Focus.

3.18.1 Range.- Rotation of the focusing knob shall bring targets from 50 meters to infinity into focus.

3.18.2 Affect on diopter setting.- The diopter setting shall not change more than 0.25 diopter, as read on the diopter scale, when the focusing knob is rotated through the range specified in 3.18.1.

3.19 Operability.

3.19.1 Disengagement of knob locks.- The direct force required to disen-gage the diopter knob and each boresight knob from the lock position at standard temperature shall be within the limits of 2.0 to 5.0 pounds (lbs). The direct force required at the extreme operating temperatures shall not exceed 9.0 lbs.

3.19.2 Torque.- The running torque required to rotate the controls listed below shall be within the limits of the values specified for standard temperature and shall not exceed the values specified for extreme operating temperatures. The boresight slip scales and the filter lever shall be manually operable at all of the above temperatures.

CONTROL	STANDARD TEMPERATURE	EXTREME OPERATING TEMPERATURE
Boresight knobs	2.0 to 5.0 in. lbs	8.0 in. lbs
Diopter knob	3.0 to 7.0 in. lbs	14.0 in. lbs
Focusing knob	3.0 to 8.0 in. lbs	15.0 in. lbs

3.20 Interior atmosphere.- The interior atmosphere of the body assembly shall contain nitrogen having a dewpoint at least as low as -25°F.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of Product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926 and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL 10513391. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

		REQUIREMENTS	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 0.65 percent defective			
101.	Collimation - Pre-vibration	3.12.1	4.5.7
102.	Vibration	3.5.4	4.5.8
103.	Cleanliness, unity power system	3.7, 3.7.1	4.5.9
104.	Image tilt, unity power	3.13.1	4.5.10
105.	Resolution, unity power	3.16.1	4.5.11
106.	Resolution, nine power system	3.16.2	4.5.12
107.	Cleanliness - Nine power system	3.7, 3.7.2	4.5.13.1

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		REQUIREMENTS	TEST PROCEDURES
108.	Flicker-movement	3.10.1	4.5.13.2
109.	Sparking	3.10.2	4.5.13.3
110.	Collimation, post vibration	3.12.1	4.5.13.4
111.	Collimation change, mounting	3.12.2	4.5.13.5
112.	Reticle pattern definition	3.11.1	4.5.13.6
113.	Reticle accuracy	3.11.2	4.5.13.7
114.	Image tilt, nine power system	3.13.2	4.5.13.8
115.	Parallelism of reticle and image	3.14	4.5.13.9
116.	Boresight movement, azimuth	3.15.1	4.5.13.10
117.	Boresight movement, elevation	3.15.1	4.5.13.11
118.	Reticle shift, azimuth and elevation	3.15.2	4.5.13.12
119.	Backlash, azimuth and elevation	3.15.3	4.5.13.13
120.	Zero setting, diopter scale	3.17.1	4.5.13.14
121.	Excursion range, diopter movement	3.17.2	4.5.13.15
122.	Backlash, diopter movement	3.17.3	4.5.13.16
123.	Focusing range	3.18.1	4.5.13.17
124.	Focus affect on diopter setting	3.18.2	4.5.13.18
125.	Force, disengagement of diopter knob	3.19.1	4.5.14
126.	Force, disengagement of boresight knobs	3.19.1	4.5.15
127.	Torque	3.19.2	4.5.16
128.	Dielectric strength	3.9	4.5.17
129.	Sealing (+60° to 90°F.)	3.8	4.5.5
130.	Interior atmosphere (+60° to 90°F.)	3.20	4.5.6

MINOR - None defined

NOTE: The tests in Table I shall be conducted at temperatures between 60° and 90°F.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Control tests.

4.3.1 General.- One body assembly shall be selected at random as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

	CONTROL TEST	REQUIREMENTS	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 - visual

4.3.2 Environmental.- Three units shall be selected at random as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

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TABLE III

CONTROL TEST	REQUIREMENTS	TEST PROCEDURES
303. Storage temperature (-65° and +160°F)	3.5.1	4.5.3
304. Extreme operating temperature (-40° and + 150°F.)	3.5.2	4.5.4

4.3.3 Control test failure.- Should any one item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by theGovernment until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents IEL 10513391.

4.4.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.2.3 Inspection equipment design requirements.

4.4.2.3.1 Sealing.- Special testing equipment to test sealing of the body assembly shall conform to the internal pressure test requirements of sheet 1 of Drawing F8565556, Pressure Tester-Two Station.

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4.4.2.3.2 Performance of the 9 power system.- Special testing equipment to test performance of the 9 power system shall consist of a collimator projector having a clear aperture of at least 7 inches; a holding device capable of positioning the periscope body with Datum Plane "A" horizontal; a mirror conforming to the dimensional and optical requirements of Dwg. D8215990, except that the special coating is not required. The necessary keyways, locating surfaces and body mounting features of the Mount Assembly 10513500 shall be incorporated into the holding device. Adjust the collimator line of sight (optical axis) horizontal and perpendicular to Datum Plane "B" in azimuth only. Position and adjust the mirror to a 45 degree angle so that the line of sight of the collimator is perpendicular to Datum Plane "A". An appropriate calibrated reticle capable of measuring the optical accuracies of the periscope body assembly shall be utilized in the collimator projector. The reticle shall contain a boresight mark 1 degree 30 minutes from the collimator line of sight (in the elevation plane), such that the reflected collimated beam from this direction makes an angle of 90 degrees plus 1 degree 30 minutes with surface Datum Plane "A". This angle is measured in a clockwise direction from Datum Plane "A" toward surface Datum Plane "B" (see Dwg. F10513391, Sheet 2). A power supply for energizing the oscillator and reticle projector of the periscope body assembly shall be part of the special testing equipment. The power supply shall be capable of supplying and maintaining the applicable voltages required in 3.6.

4.5 Test methods and procedures.

4.5.1 Image tube.- Evidence that the image intensifier tube conforms to the requirements specified in 3.2 shall be provided at the time of inspection of the body assembly. This evidence shall become part of the inspection record for the body assembly.

CAUTION: Do not expose the body assembly to direct sunlight when the image intensifier tube is energized because excessive light will damage this component.

4.5.2 High voltage oscillator.- Evidence that the oscillator conforms to the requirements specified in 3.3 shall be provided at the time of inspection of the body assembly. This evidence shall become part of the inspection record for the body assembly.

4.5.3 Extreme storage temperatures.- Testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and conditions of 3.5.1. The body assembly shall be exposed to each specified temperature extreme for a minimum of five hours. Subsequent to the temperature exposures the body assembly shall be stabilized in an ambient temperature between +60° and 90°F and subjected to the audible, tactile and visual examination of 4.5.8. The body assembly shall conform to the requirements of 3.5.1. Caution shall be exercised during this test to avoid subjecting the body assembly to thermal shock.

4.5.4 Extreme operating temperatures.- Testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.5.2. The body assembly shall be exposed to each temperature extreme specified in 3.5.2 for a minimum of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the body assembly shall

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be subjected to the tests specified in 4.5.14 through 4.5.16 inclusive. The body assembly shall conform to the operability requirements of 3.5.2 and 3.19 for each temperature extreme. Caution shall be exercised during this test to avoid subjecting the body assembly to thermal shock.

4.5.5 Sealing.- The internal pressurizing of the body assembly to test sealing, shall be performed with special testing equipment conforming to 4.4.2.3.1. Pressurize the body assembly, with dry nitrogen gas to the value specified in 3.8, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the body assembly for a period of 5 minutes ± 15 seconds. During this time period, the boresight knobs, diopter and focus knobs shall be rotated to their clockwise and counterclockwise stops. At the end of the 5 minute period there shall be no evidence of leakage, as indicated on the pressure gage of the testing equipment, to determine compliance with 3.8. The body assembly shall then be subjected to the test in 4.5.6.

4.5.6 Interior atmosphere.- A dewpoint tester shall be used to test the dewpoint of the interior atmosphere of the body assembly. Upon satisfactory completion of the sealing test, the internal pressure of the body assembly shall be reduced to 1 psig. The body assembly shall then be flushed with dry nitrogen gas. The flow of gas into the body assembly shall be controlled by a regulator that will prevent the internal pressure in the body assembly from exceeding 1 psig during the flushing operation. The flushing shall continue until the nitrogen gas bled from the body assembly has a dewpoint at least as low as minus 25°F, as measured with the dewpoint tester. The body assembly shall then be sealed. The internal atmosphere of the body assembly shall conform to the requirements of 3.20.

4.5.7 Collimation - pre-vibration.- Prior to the vibration test, the body assembly shall be positioned as specified in 4.5.13. Energize the image tube and reticle projector. While observing through the body assembly eyepiece, establish coincidence of the reticle boresight mark with a point on the projected target image, utilizing the boresight knobs. Record the position of the boresight knobs with respect to the body housing, and the boresight mark with respect to the target image. Remove the body assembly from the holding device. The body assembly shall then be subjected to the tests specified in 4.5.8.

4.5.8 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.5.4, shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in conformance with the "Test Facilities" requirements of MIL-F-13926. The body assembly shall be positioned to the vibrating equipment in the manner in which it is to be used, by means of an adapter containing the necessary keyways, locating surfaces and body mounting features of the Mount Assembly, Drawing F10513500. The body assembly shall be vibrated in a vertical plane for a period of 5 minutes ± 15 seconds. Subsequent to vibration the body assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.5.4, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts, such as (boresight knobs, focus and diopter knobs, etc). An audible examination shall be made of the body assembly for loose internal parts or broken optical compenents, such as (lenses, prisms reticle, image intensifier tube, oscillator, etc). Anyone of the possible defects defined above shall be cause to

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consider a body assembly defective, unless separate defects-per-hundred type sampling plans are established and approved by the Government as part of the suppliers inspection system. Body assemblies which have passed the above examinations shall be subjected to the tests of 4.5.9 to 4.5.16 inclusive to determine overall compliance with 3.5.4.

4.5.9 Cleanliness, Unity-power system.- The visual system of the body assembly shall be inspected for cleanliness in accordance with MIL-O-13830, and shall meet the requirements specified in 3.7 and 3.7.1.

4.5.10 Image tilt, unity power.- Special testing equipment consisting of a holding device capable of positioning surface Datum Plane "A" of the body assembly vertical, a calibrated wall target in conjunction with a plumb line, and a level (horizontal) surface for positioning an auxiliary telescope at the exit window of the body assembly shall be utilized for this test. A calibrated auxiliary telescope at least 3 power (3X) containing a horizontal and vertical reticle line shall also be utilized for this test. Secure the body assembly to the holding device and direct the line of sight of the body assembly to the plumb line wall target. Position the telescope on the level surface at the exit window aperture and plumb the vertical reticle line of the telescope. Observe through the telescope and establish one extremity of the telescope vertical reticle line coincident to the vertical target plumb line image. The image of the target plumb line shall be parallel to the vertical reticle line of the auxiliary telescope within the tolerance specified in 3.13.1, as measured on the target.

4.5.11 Resolution, Unity power system.- Perform this test with the body assembly positioned as in 4.5.10, utilizing a resolution target and an auxiliary telescope of at least 10 power (10X). The resolution target shall be a resolving power chart on a wall target, representing the angular subtense for seconds of arc specified in 3.16.1. Construction of the resolving power chart shall be as specified in the "Resolution test" of MIL-O-13830 and shall contain four line sets as shown in Figure 1 therein. The target shall be illuminated with white light. Position the telescope at the exit window aperture, observe through the telescope and scan the longitudinal axis of the optical system. All four meridians of the resolving power chart shall have the detectable line structure and proper line count. The resolution shall meet the requirements specified in 3.16.1.

4.5.12 Resolution, nine power system.- Conduct this test utilizing a resolving power chart on a wall target representing the angular subtense for the minutes of arc specified in 3.16.2. Construction, location and illumination of the resolving power chart shall be as in 4.5.11. Position the body assembly as in 4.5.10, and energize the oscillator. After the image tube has stabilized (see 4.5.13) adjust the diopter knob for best focus of the phosphor grain on the screen of the image tube. View the target image through the body assembly eyepiece with the unaided eye. Obtain best focus of the numeral in the center of the resolving power chart by means of the focusing knob of the body assembly. There shall be no further adjustment of the focusing knob during this test. At this point of focus of the nine power system, all four meridians of the resolving power chart shall have the detectable line structure and line count to conform to the requirements of 3.16.2.

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4.5.13 Performance, Nine-power system.- Conduct the following tests with the body assembly positioned on special testing equipment conforming to 4.4.2.3.2. Apply the operating voltages specified in 3.6 to the reticle projector and the oscillator of the body assembly. The reticle projector and oscillator shall be energized unless otherwise specified. The image tube shall be permitted a stabilization period of 15 seconds prior to conducting optical tests of the nine power system.

4.5.13.1 Cleanliness, nine power system.- Visually inspect the optical system and image screen in the body assembly for cleanliness, while the line of sight of the body assembly is directed to a plain light backdrop that will cause uniform illumination of the image tube screen. Cleanliness of the body assembly shall conform to the requirements of 3.7 and 3.7.1.

4.5.13.2 Flicker movement.- Perform this test with the reticle projector switch in the "OFF" position. Position an auxiliary telescope at the eyepiece of the body assembly. The auxiliary telescope shall contain a crossline reticle and be capable of focusing on the screen of the image tube in the body assembly. Observe through the auxiliary telescope and obtain coincidence of the horizontal and vertical reticle line and the image of the collimator reticle pattern lines on the screen of the image tube. Observe the target image for a time period of not less than one minute. During this one minute time period there shall be no visible flicker or movement of the image as specified in 3.10.1. Image movement shall be determined by means of the target image with respect to the auxiliary telescope reticle pattern. The body shall then be subjected to the test specified in 4.5.13.3.

4.5.13.3 Sparking.- Perform this test subsequent to, and utilizing the procedure specified in 4.5.13.2. During the one minute period of observation there shall be no visible or audio evidence of sparking as specified in 3.10.2. Upon completion of this test return the reticle projector switch to the "ON" position.

4.5.13.4 Collimation change-due to vibration.- Rotate the boresight knobs to the position recorded in 4.5.7. Observe through the body assembly eyepiece. Any adjustment of the boresight knobs to obtain coincidence of the reticle boresight mark with the recorded point on the target image established in 4.5.7, shall not exceed the tolerance specified in 3.12.1, as measured on the boresight knobs. Record the new collimation setting of the boresight knobs, if necessary, for use in the test in 4.5.13.5.

4.5.13.5 Collimation change, mounting.- Perform this test in sequence to the test in 4.5.13.4. Remove the body assembly from the holding device. Repeated installation of the body assembly on the holding device (a minimum of three times) shall be performed. The body shall be inspected after each installation for repeatability of collimation. The body assembly shall meet the collimation requirements specified in 3.12.2, as measured on the boresight knobs.

4.5.13.6 Reticle pattern definition.- Perform this test in accordance with MIL-O-13830. This test shall meet the requirements specified in 3.11 while viewing through the body assembly eyepiece.

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4.5.15 Force, disengagement of boresight knobs.- Perform this test specified in 4.5.14. The disengagement of the boresight knobs shall meet the requirements specified in 3.19.1.

4.5.16 Torque.- The torque required to rotate the diopter knob, boresight knobs and focusing knob shall be measured with standard measuring equipment affixed to the knobs. The measured torque of the knobs shall not be less than or exceed the values specified in 3.19.2, at the temperature indicated. The boresight slip scales and filter lever shall be subjected to a tactile examination, and shall conform to the requirements specified in 3.19.2 at the temperatures indicated.

4.5.17 Dielectric strength.- The body assembly shall be tested in accordance with Method 301 of MIL-STD-202, utilizing standard measuring equipment. Dielectric strength shall be in conformance with the requirements of 3.9. This is an in-process test that is performed with the image tube and oscillator removed from the circuitry. The inspection records for each body assembly shall indicate the results of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10513391. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Tank Periscope Body Assembly 10513391 is a component of the XM44 Tank Periscope. The XM44 Tank Periscope is primarily intended for use as the gunner's sighting and observing system in the XM551 Vehicle.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level or preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).

6.3 Image intensifier tube.- The detailed requirements for the image intensifier tube are specified in "Purchase Description for Image Intensifier Tube Assembly, (25 Millimeter) Type 8586 (U)" dated 8 July 1964. This purchase description is a classified document.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A443Y

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